

Asset Markets and Liquidity

18.1 CONCEPTS

The liquidity of an asset is the ease with which it can be converted into cash when required, or conversely, the ease with which it can be bought for cash. In both cases the focus is on such a conversion taking place without affecting market prices too much. This definition explains why the word liquidity is used in this context in the first place: more generally, liquidity refers to the cash or cash-like instruments held by an entity. Market liquidity then refers to the general liquidity conditions of the assets in that market, e.g., the market for US Treasuries or Japanese municipal bonds.

Liquidity is not constant. While a typical government bond in Europe will see roughly unchanged trading activity throughout its life, turnover in a usual corporate bond will drop by around 90% within the first six months of its life. US Treasury bonds also show declining activity in older securities while in the JGB market liquidity differs across the various types of JGB. Liquidity during a usual trading day tends to be higher in the morning hours than in the afternoon and important economic data releases or other news events can drain liquidity while the market digests the new information.

One aspect of liquidity is tied to a search problem: A seller of an asset needs to find a willing buyer for that specific asset at the current time at a price that is at or near the indications available in the market. Simple combinatorics imply that this search problem is easier when the number of assets is smaller, or the number of potential buyers is larger. Issuers can therefore aid liquidity by restricting the number of securities they issue and produce larger volume securities instead. This is one reason for the relative scarcity of very long-dated bonds: An issuer who issues a new 100 year bond every year will eventually have 100 bonds outstanding and so split market liquidity across so many securities while the sum of the outstanding amounts of these bonds equals the total debt of the issuer. If the same issuer issues 10-year bonds instead, each one can be 10 times larger and is likely to be more liquid as a result¹. Issuers can also make it easier for a broad range of investors to participate in the market by avoiding artificial constraints related to settlement and custody arrangements.

¹The downside for the issuer is that shorter bonds require more frequent refinancing and so expose the issuer more to interest rate volatility.

The conversion between assets and cash can happen in a variety of ways, two extremes of which are public exchanges and market makers. Public exchanges are sometimes called central limit order books, or CLOBs in reference to the most common technical structure of these venues: Exchange participants send orders to the exchange that specify the asset, the side (buy or sell), a quantity, and a limit price which is the highest purchase price, or the lowest sale price, that the participant will accept. These orders are collected in a central book, and matched up against each other whenever possible. Participants have the option to modify or rescind orders in response to market news.

Market makers, in contrast, hold themselves out to provide prices to their clients in what is known as the over-the-counter (OTC) market, regardless of whether there is an intermediate offsetting trading interest from another client. Two protocols are used for such transactions. The most common by far is the request for quote (RFQ) protocol where a client requests a firm price for a given size and transaction side in a specific security, and the market maker responds with a price, or, less commonly, with a refusal to trade. Trading can take place on electronic platforms or via phone or chat (both the latter are referred to as ‘voice’ trading). The alternative streaming pricing protocol is one where dealers continuously post executable prices on some electronic platforms and clients can choose to execute at these prices up to a certain maximum size. Regulators prefer the streaming protocol for transparency but in practice dealers have little economic incentive to offer good pricing on such services. It is an essential component of market making that firms offering such a service are willing to warehouse securities for some time until they are able to lay off the positions to some other market participant.

The search problem mentioned above is approached in different ways by exchanges and market making. Exchanges aim for the concentration of a large number of investors in the same venue. Market making essentially eliminates the search problem by producing willing counterparties, albeit at a cost (the bid–offer spread charged by the market maker and any information advantage gained from being in this business). To make this possible, market makers keep their positions secret (at least from the wider market) to prevent information leakage.

While public exchanges have been ubiquitous for a long time, their existence in a capitalist economy is worth a second thought. Exchanges are shared utilities of their participants but these participants are by nature competitors. Exchange trading is an obvious zero-sum game where every change in price of a transaction creates a different redistribution of wealth between the two parties that transact. There are two possible explanations why exchanges form despite this natural antagonism of their users. The first is that exchanges divide the total set of investors into one of insiders (exchange members) and outsiders (everyone else). For insiders, any disadvantage created by the succour given to the other insiders is outweighed by the advantages enjoyed collectively by the insiders vis-à-vis the outsiders who do not have access to the trading venue. In economic terms, exchange members form a cartel and enjoy economic rents. An example of such an advantage was the existence of fixed (i.e. non-competitive) minimum commissions charged by members of the London Stock Exchange to the investing public before the ‘Big Bang’ reforms implemented on 27 October 1986 [59]. A second potential explanation is that the reduction of search costs and collective regulation is a sufficiently valuable public good for the exchange members to offset the disadvantage

of supporting one's competitors. Rather than being concerned with the benefits their activity bestows on their competitors, exchange members value the reduction in potential risks to themselves created by the ability to control or lay off risks to other firms. This type of outcome is an example of a network effect². This network effect could be achieved by other means, however. The EuroMTS trading platform is an inter-dealer trading venue where dealers post firm prices to each other. The existence of the platform allows dealers to exit unfavourable positions, but the bid-offer spreads seen on this platform are wider than the general market. The MTS platform is therefore not the optimal pricing venue for dealer trading, and is not designed for that purpose. In any case, dealer has a strong incentive to sell a bond to a client at the market offer rather than at the MTS bid to another dealer. As a result, while MTS provides good price discovery, the actual execution activity on the platform is dwarfed by the overall market volumes.

Public exchanges and OTC markets compete for client trading activity. Investors sending orders to an exchange can look for execution at a transparent venue for comparatively low fees but by sending in their order they make public what trade they are looking to execute. The information disclosure alone can lead to an adverse move in market prices; this effect is called information leakage. In addition, while information leakage is virtually certain to occur, there may be no opposing order in the CLOB that would be match the trading interest so exposed, at least not in a short period of time and near market levels prevailing when the initial order was posted. In this case, the investor has disclosed information but not managed to execute the trade. Sending the order to a market maker in the OTC market discloses the trade interest only to that counterparty but market makers will seek to generate profits by seeking to quote low bid and high offer prices. Investors can try to restrict such behaviour by sending the same pricing request to multiple market makers at the same time ('in competition'³). However, market making is overall a business that not only covers the cost of the resources involved but is also profitable. This observation implies that users of the OTC market are paying for the provision of liquidity. The economic trade-off between exchange and OTC execution is therefore essentially tied to search costs. When many investors trade a relatively small number of distinct assets, public exchanges are likely to be an efficient execution venue. When there are too many similar but not fully fungible assets, trading interest is dispersed across them and OTC execution is more likely to be more efficient. This explains why blue-chip equities and bond futures contracts trade comfortably on exchanges, while bonds and small-cap equities trade mostly in the OTC market.

The competition between public exchanges and OTC markets suggests that it is unlikely that market makers are able to extract rents in the same way as exchange members could. Even if one were to argue that market making is not sufficiently competitive,

²Cities exhibit similar network effects in that traders of related crafts congregate in the same areas because the benefit of finding customers easily outweighs the downside of always being in close competition. Street names like Bread Street in London and areas like Takaracho in Tokyo are reminders of these times.

³It is considered polite to disclose the presence of competition to the market maker. Many investors have regulatory obligations to trade in competition ('in comp') involving at least three market makers.

the market could always move to exchanges if trading there was cheaper. Given that the majority of bonds are listed to satisfy formal criteria, trading them on exchange is to all extents already possible but apparently not economical⁴.

Exchange trading and market making are not mutually exclusive. Some exchanges occasionally operate compensation models for some products where exchange participants who commit to providing continuous bid and offer prices share in the exchange fees paid by other users who trade on these prices. This is typically done in order to generate an initial trading interest in new products and continues until a natural market is established⁵. In other instances, specialist trading firms (known in the US as principal trading firms, or PTFs) act like market makers in normal times without being formally recognised as such. This does, however, also imply that such firms will not make prices when trading conditions become less favourable for them.

Recently, alternative forms of bridging the gap between public exchanges and market making have emerged. One is algorithmic trading, which is a catch-all term for strategies that execute trades on public exchanges without disclosing the true trade interest to the wider market. When such strategies are used to manage larger trades without disturbing market prices too much, this is generally not viewed as negative. However, the same types of algorithm can also be used to mislead the market about trade interest in order to move prices. Such strategies, known as spoofing, are considered to be market abuse. The other alternative is dark pools, which operate like public exchanges but without a public disclosure of orders⁶. Unlike public exchanges ('lit' markets in this context), dark pools disclose trade interests only when orders can actually be matched. Financial firms that have traditionally been market makers have set up dark pools to augment their market making business, which requires committing balance sheet to the warehousing of positions, with what is essentially a match-making service for clients. For regulators, dark pools detract from the goal of achieving more transparency in financial markets.

From an investor's point of view, the picture is not perfectly clear. Each investor would like to know about each of the other investors' trading conditions while having an interest to conceal her or his own trades from the wider market to prevent information leakage. Both goals cannot be achieved at the same time.

⁴The German Bundesbank provides a service which consists of posting prices on German public exchanges based on the liquidity of the OTC market. If exchanges were a superior execution venue, this service would not need to exist.

⁵Such arrangements are not without complications. The exchange is in effect spending funds of exchange users on the creation of trading activity that would not otherwise exist. This may be seen as misleading these users as to the actual liquidity in the products involved. As long as such arrangements are properly disclosed and limited in duration regulators do not normally intervene. However, there has been at least one instance, involving the 10Y OAT contract on Matif in 2000, where regulators did step in to stop what was perceived as an artificial inflation of trading activity [55]. In this case French banks had set up a 'Matif intervention bancaire', inevitably known as the 'Men in black' by its acronym, to generate extra trading activity.

⁶A dark pool is therefore comparable to a broker market, but available to a wider public.

18.2 LIQUIDITY MEASUREMENT

The definition of liquidity has some obvious conceptual problems. There are multiple venues in which assets can be traded and each of them may lead to different assessments of liquidity. Not every investor has access to every public exchange, every dark pool, or is being serviced by every market maker. Liquidity therefore is dependent on the investor, not just on the asset. Practically, most liquidity indicators are therefore measured for a given execution venue and therefore apply to a certain class of investors. This means, however, that contagion across trading venues is by construction not reflected and liquidity may be understated as a result.

Moreover, measuring liquidity relies on one of two counterfactual assumptions, namely ‘At what price could a trade take place (while it does not)?’ or ‘What would the market price be if a trade had not taken place?’ These two counterfactuals open two problems: How reliable are price quotations when they are not provided with a firm intention to trade; and, How can price movements caused by actual trades be separated from price movements caused by other factors?

The first concern about the reliability of quotes is related to advertising. Market makers have an incentive to show very competitive quotes on an indicative basis in order to attract requests for firm quotes. Meanwhile, algorithmic trading algorithms operating on public exchanges also are designed to suggest that trading opportunities are available at or near current market prices. In either case, the intent is to incentivise investors to reveal true trade interest. At the time such interest is revealed, the market picture might change. A market maker may refuse to quote on a firm basis, or an algorithm may choose to cancel an order. In either case, the indicated market prices can be unavailable for actual execution.

The second concern above is reminiscent of the Heisenberg Uncertainty Principle in physics. Prices are fundamentally a stock measure in the sense that they are point-in-time assessments of the value of an asset. Trading volumes are, on the other hand, flow measures that reflect the amount of an asset that has changed hands over a certain period of time. To measure flow precisely, it needs to be observed over a certain period of time, while over that period of time other factors, such as economic data releases, can affect prices. This means that price measurement becomes more uncertain if flow is observed over longer time intervals.

Another important concern is about the very notion of trading in the context of defining liquidity (see Figure 18.1). One can treat each transaction as an event by itself

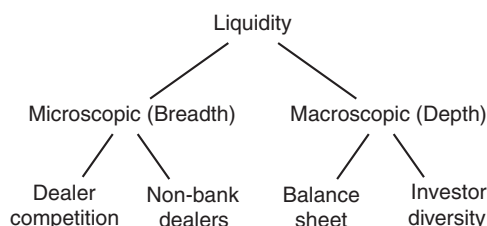


FIGURE 18.1 Multiple aspects of liquidity.

and define liquidity related only to this single trade instance. This microscopic aspect, also called market breadth, is largely related to the competitive environment between market makers and the willingness of non-bank dealers to respond to trade orders or quote requests. In a wider context, however, the totality of transactions performed in a given period will have some common elements, such as dis-investment from a certain subset of securities. This creates a correlation between individual transactions that will be reflected in market pricing. This is the macroscopic aspect of liquidity, called market depth. The ability of the market to absorb correlated flow is related to the aggregate risk absorption capacity of liquidity providers, including non-dealers with active trading operations. The number or competitive situation of liquidity providers is less important in this context than the capacity to hold positions in the face of market volatility.

In this sense, macroscopic liquidity is harder to observe because it is only meaningful when trading patterns are less diversifiable. From a market stability point of view, this is somewhat disconcerting because it means that a lack of market depth will only be observed when market depth is needed. Simply relying on microscopic liquidity measures to gauge market depth is however not an option given the arguments made above.

18.2.1 Taxonomy of liquidity measures

There is a multitude of quantitative liquidity indicators used in the market and this requires some form of taxonomy. One approach is to classify indicators by the type of information used, as done in Table 18.1.

What may be surprising is the inclusion of measures that do not rely on trade information at all. The thinking behind such measures is that a liquid market has secondary properties, such as the absence of obvious arbitrage opportunities. Illiquidity can therefore manifest itself through the appearance of apparent mis-pricing.

Each type of measure suffers from particular problems. However, in many instances the trends shown by different indicators tend to move in the same direction [36]. When they do not, their divergence may reveal deeper information about changes in market conditions, such as investor perception of trading conditions as opposed to current liquidity.

Pre-trade measures rely on price quotations that may not be firm, or could be withdrawn before they can be executed on. Liquidity apparent in such prices may also disappear at any time. Most aggregated indicative screen price bid-offer spreads are substantially wider than what could be achieved in actual trading although individual dealers may window-dress their own indications to be tighter than what they are

TABLE 18.1 Liquidity measures by type of information used

Pre-trade information	Post-trade information	No-trade information
Bid-offer spreads	Trade impact	Spline spread dispersion
Order book depth	Trade volumes	Futures dislocation
Repo rates	Quote dispersion	
	Time to quote	
	Trade sizes	

actually willing to quote in firm inquiries. At the same time, algorithmic traders may be hiding the full extent of their trade interest from public order books, so liquidity might actually be better than the visible order book suggests. Last but not least, many markets have minimum price gradations. This introduces quantisation noise into any price measures. Other pre-trade measures are the frequency of price updates (less liquid markets will see fewer updates) but the increasing prevalence of machine pricing means that prices may be updated by an algorithm with no reference to actual trading activity. Commercial providers produce related indicators based on actually transacted prices relative to indicative mid prices. That being said, sample bias can be a significant problem for such measures because in stressed liquidity situation there will be no bid and offer prices for less liquid securities, and trading will be concentrated in the most liquid securities.

Post-trade indicators are based on actual trades, rather than potentially misleading indications. Typically, they relate the price changes observed in a given time period to the observed trade volumes. However, they too may be misleading for three reasons. First, not every trade is observable by every market participant so any measurement will be based on a potentially biased subset of trades that could be observed. Second, the subset of trades that have taken place is itself a biased sample of trades that *could* take place, namely that subset that was deemed both necessary and executable at the time. To the extent that portfolio managers make trading decisions based on their judgement of relative ease of execution (which is incumbent on good portfolio managers), post-trade measures of liquidity may be biased away from illiquid securities. Third, measures related to the price impact of trade volumes additionally may be skewed by the amount of market-moving information, including trades in related markets, that emerged during the observation period. Market prices may move by large amounts even in the absence of any trading when new information arrives. Other post-trade indicators are trade size, number and volume of trades, and so on. Such indicators are also not without problems. A dormant market may burst into activity not just due to improved liquidity, but also because investors feel the need to enter or exit that market.

An interesting post-trade indicator for active trading clients is the dispersion of prices observed in competitive requests for quotes from multiple counterparties. In a liquid market, dealers can be expected to be well aware of the market-clearing price and willing to quote close to it. In a less liquid market, dealers will be both uncertain about the prevailing price level, and more inclined to adjust prices to their current inventory. Given that dealers sometimes use off-market prices to signal an unwillingness to trade, care needs to be taken to disregard such unrealistic responses. As other post-trade indicators, this type of measure suffers from sample bias as portfolio managers select the trades that appear executable, while the execution traders select the counterparties that are likely to offer reasonable prices.

The practical definition of indirect indicators relies on a reasonably reliable implementation of the arbitrage concept. An arbitrage opportunity is always a theoretical concept and may not be present in practice, or the identification of opportunities may be flawed. For instance, spline spreads calibrated on spot market prices may suggest over- and under-valued securities. However, the conditions for borrowing the securities identified as overpriced for short-selling, or the cost of funding long positions in the apparently underpriced securities, may make it impossible to trade on this perceived

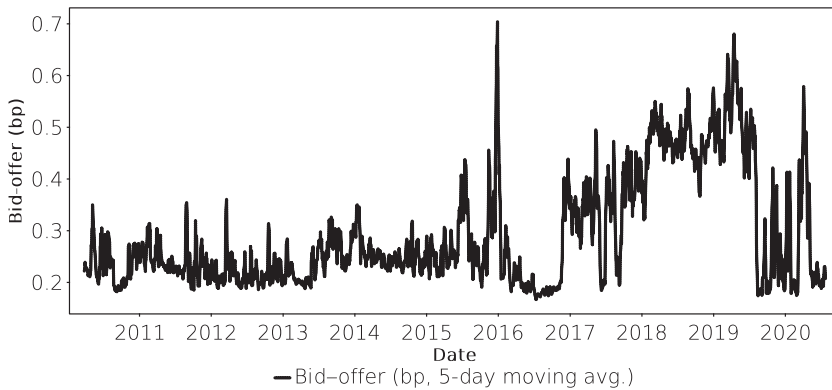


FIGURE 18.2 Bid-offer spreads (in yield basis points) of nominal US Treasuries with remaining maturities of 7–15 years, calculated from data obtained from treasurydirect.gov. Note that the convention to quote Treasuries in 1/256th steps creates an effective lower bound on the bid-offer spread equivalent to around 0.04 bps in this maturity range. European bond markets are less liquid but are quoted in price steps of 1/1,000th (0.1 cents). Source: Data obtained from treasurydirect.gov.

opportunity. Moreover, the removal of arbitrage opportunities through trades that take advantage of them can only occur to the extent that market participants judge liquidity conditions to be sufficient. Since such judgements can take time to adjust, and can be disturbed by news flow, the link to actual liquidity conditions can be tenuous at times. On the other hand, indirect indicators are not biased by advertisement effect of indicative prices, and less biased by trade selection than the post-trade indicators⁷. In this respect, indirect indicators may be a better measure of macroscopic liquidity, with the caveat mentioned above that, in the absence of undiversified flow, such indicators will not signal a deterioration of liquidity conditions.

To sum up, each type of liquidity indicator has advantages and disadvantages. Indirect indicators may be better at reflecting problems in the macroscopic liquidity, while the pre- and post-trade indicators reflect microscopic liquidity developments. This means that rather than relying on a single indicator, it is useful to monitor a set of diverse indicators and pay as much attention to the divergences observed as to the commonalities.

18.3 EXAMPLES

The production of pre- and post-trade bid-offer indicators is structurally fairly simple as long as the relevant data is available. Figure 18.2 shows an example of average bid-offer spread data for US Treasuries.

As an example of indirect liquidity indicators, Figure 18.3 shows average absolute spline spreads against a Nelson-Siegel spline. The production of such indicators is more

⁷Indeed one could argue that indirect indicators reflect the cost of trades that are not done, instead of the trades that actually take place.

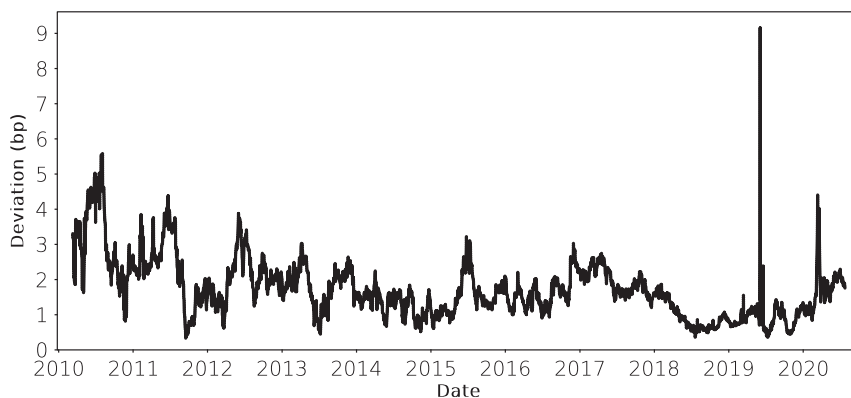


FIGURE 18.3 Average absolute deviations from a Nelson-Siegel spline for US Treasuries with remaining maturities of 7–15 years, calculated from data obtained from [treasurydirect.gov](https://www.treasurydirect.gov). The spline is fitted to maturities from 1 to 30 years and spreads are re-based to average 0 for all bonds in the daily sample. Source: Data obtained from [treasurydirect.gov](https://www.treasurydirect.gov).

complex than for Figure 18.2. First, a fair value model has to be fitted to the observed market prices. Here, this model has been fitted to the entire curve. The average absolute yield deviation from fair value is then calculated for a subset of bonds so liquidity conditions in different parts of the curve can be assessed independently. To remove the impact of sectoral richness or cheapness, the spreads are first re-centred to a zero average.

Both measures show low liquidity (high indicator values) around the latter stage of the financial crisis and around the time of the ‘taper tantrum’ of 2013. What is more interesting is the divergence of both indicators in 2015/16. While the microscopic bid–offer spread indicator shows no meaningful deterioration, spline spreads show a greater dispersion. Given that this period coincided with the activation of restrictions on trading introduced by the Dodd–Frank Act, perhaps there could be a connection to lowered provision of dealer balance sheets, particularly in the repo market.

Changing liquidity conditions are often correlated across markets because operating constraints affect dealers in similar ways.

Figure 18.4 shows identically calculated liquidity indicators across the US, Germany and Japan during the worst phase of the market disruption caused by the Covid-19 outbreak in 2020. All three markets show substantial strains going into March. Factors contributing to this development were funding difficulties for some market participants and pressure to reduce risk positions in the face of heightened volatility. On 15 March, the Federal Reserve announced significant measures which were in part aimed at supporting Treasury market functioning, including a market-neutral approach to purchases and extended repo lending of the SOMA portfolio. These measures supported a notable improvement in the US market. The ECB’s public-sector purchase programme (PSPP) had already been implemented in a market-neutral fashion since its inception in 2015 which partially explains why the liquidity situation in the Bund market did not deteriorate to the same extent as in the US. Still, the ECB added an additional purchase

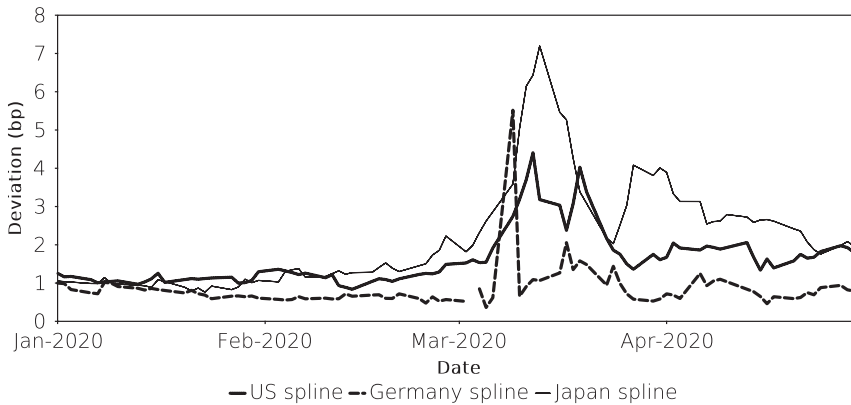


FIGURE 18.4 Spline liquidity indicators for US Treasuries, German Bunds and JGB calculated in the same way as in Figure 18.3 during the worst phase of the Corona virus in 2020. Sources: Data from TreasuryDirect, Bundesbank and JSDA.

envelope to PSPP and on 18 March launched the Pandemic Emergency Purchase Programme and market liquidity also improved.

18.4 LIQUIDITY PREMIUM

The total return of an investment depends on the transaction costs related to entering and leaving the position. Simplistically, the return of a bond is given by:

$$\text{Return} = [\text{Coupon income}] + [\text{Capital gain}] - [\text{Bid} - \text{offer spread}] \quad (18.1)$$

The items in this equation have different time dependencies. If a bond is bought and sold again immediately, then the first two terms will be zero while the third term, which will usually not be zero, creates a loss. Over very long periods, the first two terms should dominate this equation.

From the perspective of a portfolio manager, the return of the entire portfolio is the sum of such terms for each individual asset held. Because usual portfolios require trading, either to respond to inflow and outflows, or to changes in outlook, trading costs, i.e., the third term in this equation, are of great importance.

Some assets are known to be tradeable with lower transaction costs because they have a deeper market, or because market trading is concentrated in these securities. If that is the case, then a portfolio manager considering the equation above for two securities with different transaction costs (third term) will have to consider an expected holding period to weigh the first two terms against the third when comparing the securities. For a short holding period, transaction costs are relatively more important, and a more liquid but lower yielding bond may be more attractive than a higher yielding, but less liquid one. Lower transaction costs can therefore generate a yield discount, or price premium, known as the liquidity premium. In some markets, and notably the

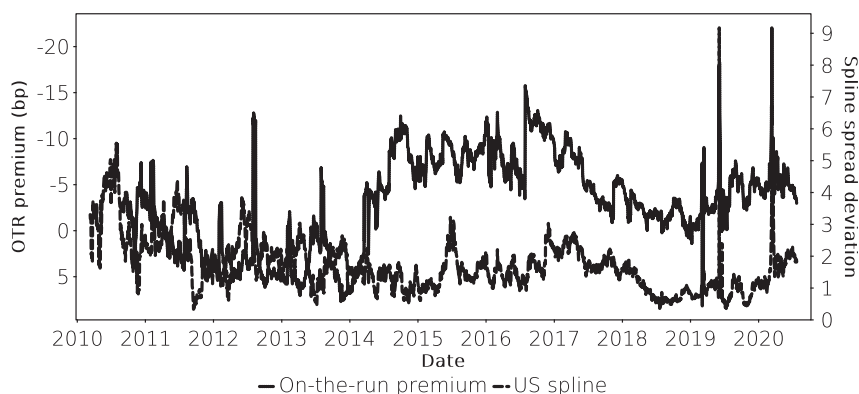


FIGURE 18.5 On-the-run premium and spline spread deviation measures of US Treasury market liquidity. Note that the spline spread deviation is plotted on the inverted right axis so that downwards movements in both series mean a deterioration in conditions. The impact of the repo squeeze in the summer of 2019, and the onset of the Corona pandemic in 2020 are clearly identifiable. Ten-day averages. Source: Data from TreasuryDirect.

US Treasury market, the most liquid securities trade at lower yields compared to more liquid ones.

This relationship between liquidity and premium is not static but changes with perceived holding periods. If investors do not see an immediate need to adjust their portfolio structures, then it would be irrational to accept the lower income associated with the liquidity premium. Conversely, in times of higher market uncertainty, there can always be a need for rapid portfolio adjustment, and liquidity premia increase. The liquidity premium is therefore in itself a measure of market liquidity, and can be used to monitor market developments.

Figure 18.5 shows a comparison of the spline-based liquidity measure discussed above and a liquidity premium measure defined simply as the spline spread of the most liquid US Treasury, the on-the-run 10Y. The two show a strong co-movement⁸ but are not identical. The different evolution is related to the conceptual difference between the measures. The spline spread dispersion is largely a measure of potential arbitrage opportunities, and a higher value means that for a larger set of securities investors do not have the means or the confidence to use such opportunities. The spline spread of the on-the-run need not represent an arbitrage opportunity because a higher liquidity premium may be warranted by market volatility. As such, the liquidity premium measures liquidity conditions via the price investors are willing to obtain liquid assets.

Many market participants use simpler measures of the on-the-run premium, such as the spread between the on-the-run and a particular off-the-run, such as the second or third youngest security of the same initial maturity. Such measures are easier

⁸Arguably, the spline spread of the on-the-run is part of the average used to calculate the total spline spread deviation. However, its weight is not large enough to dominate the volatility of the average.

to construct than the spline-based measure here but require a trade-off. The closer the on-the-run and off-the-run are in maturity, the more similar they are in liquidity and the smaller and more volatile the spread observed between them. To get a more meaningful measure, one should use older comparables to the on-the-run, but that means a larger difference in maturity dates. This pollutes the liquidity premium measure with curve movements, such as simple steepening/flattening moves. In addition, larger differences in issuance date can mean larger differences in coupon, so that yield spread comparisons are compromised by coupon effects. Both effects can be mitigated by comparing not yields but spline spreads between on-the-run and off-the-run. If one is prepared to go so far, then just using the on-the-run spline spread is simpler. It has the added advantage that idiosyncratic movements of any one off-the-run do not affect the liquidity measure.

The German and Japanese markets do not show the same extent of on-the-run premia because price discovery and duration hedging in these markets takes place in futures contracts. This difference in market structure means that the liquidity advantage of an on-the-run security is not as significant, and there is therefore no associated premium. Technically, the most liquid instruments are the futures' cheapest-to-deliver bonds but that does not always lead to a premium. Instead, when markets exhibit selling pressure, this may first lead to a cheapening of futures contracts, and thereby the CTDs, relative to their respective bond curves.

18.5 LIQUIDITY AND VOLATILITY

Because realistic market prices are usually only observable through trading, liquidity can affect price volatility through an effect known as bid-ask bounce. The mechanics of this effect are shown in Figure 18.6.

The assumption is that mid prices, which reflect the economic value of the instrument given all public known information, follow a stochastic process. As investors buy

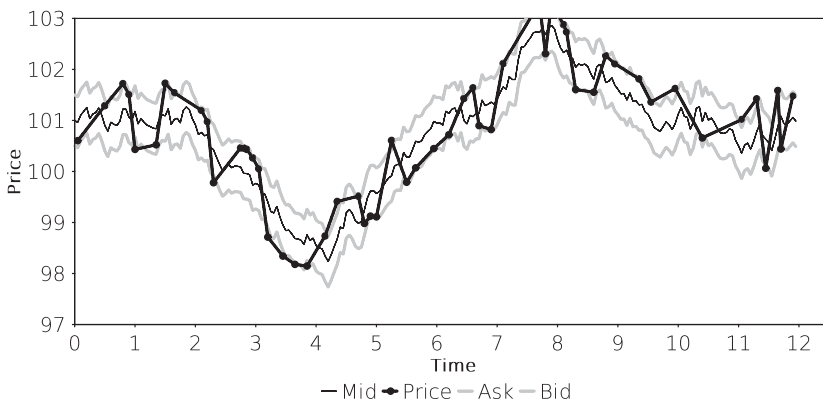


FIGURE 18.6 Observed price evolution for an illiquid instrument. As trading (dots) occasionally switches from the bid to the ask side of the market, prices appear to jump while actual mid price has much lower volatility.

and sell the instrument, however, they trade on ask or bid prices, respectively, that also reflect the liquidity of the instrument. The evolution of traded prices therefore incorporates not only the volatility of the mid price, but also the bid–offer spread. If the bid–ask spread is wide, this second volatility component can dominate the underlying volatility of the mid price.

In practice, there is no mid price that would be known to market makers but unknown to investors. Instead, one can envisage a price at which the market would clear if there was an equal amount of buying and selling interest in the market and consider that to be the mid price. At any given time, however, buying or selling interest will dominate due to the emergence of economic news, or structural factors such as index rebalancing events. Prices then need to move so much as to entice erstwhile buyers to stop buying, or erstwhile neutral investors to consider selling, for instance. This would be equivalent to the transition towards the ask price in Figure 18.6. In this case, the change in flow direction would, via this effective mid–ask or mid–bid transition, generate an apparent price volatility that would not be present in a more liquid market.